

CHAIN-SPECIFIC LIPID DYNAMICS REVEAL AN ASYMMETRIC SKIN BARRIER MATRIX

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ABSTRACT

The skin barrier, the stratum corneum is the body's first line of defence against dehydration and environmental exposure. Its exceptional barrier function is usually attributed to a dense, low-permeability lipid matrix of approximately equimolar ceramides, free fatty acids, and cholesterol, which appears ordered and largely crystalline by X-ray diffraction. Yet this static view does not fully explain skin elasticity, swelling, enzymatic activity, or the incorporation of antimicrobial peptides, suggesting that alternative asymmetric lipid organization and dynamics may exist.

Therefore, I used solid-state ²H NMR spectroscopy to quantify the molecular dynamics of deuterated skin barrier lipids and chain segments.

Our results reveal that the skin barrier lipid matrix is not dynamically homogeneous. The sphingosine chain of Cer[NS] and cholesterol display fluid-like mobility, whereas the ω -linked polyunsaturated linoleate chain of ultralong Cer[EOS] undergoes isotropic motion. In contrast, saturated lignoceric acid and the acyl chain of Cer[NS] remain predominantly orthorhombic/rigid, consistent with an extended Cer[NS] conformation not observed in phospholipid membranes. These findings support asymmetric molecular segregation of lipid and chain-segment dynamics.

To determine whether unsaturation in ω -linked chain of Cer[EOS] is required the native long-periodicity phase (LPP) barrier, we compared ultralong Cer[EOS] variants carrying native linoleate (2 double bonds), oleate (1 double bond), or saturated stearate (0 double bonds). Only unsaturated ω -linked chains preserved the isotropic mobility required for formation of the native LPP barrier. Together, these results suggest that asymmetric organization and chain-specific lipid mobility enable a more functionally diverse barrier than an exclusively rigid and symmetric lipid assembly could provide.

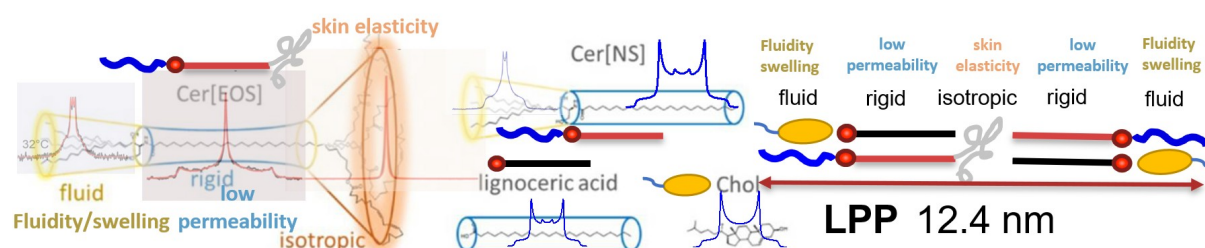


Fig: ²H NMR spectra overlaid with structural models illustrate segment-specific lipid dynamics and their contribution to a functionally diverse, asymmetric native skin barrier.